Assumptions — Read This First

The following assumptions were made while writing this guide. If any of them are wrong, the affected steps are noted so you can adjust.

1. **Multiple versions of this process were provided; this guide follows the newest one.** Six files were provided. Two of them (`title_ii_2024.R` and `title_ii_2025.R`) are complete versions of the same annual process, and four smaller files are pieces of an even older cycle of the same report. This guide is based entirely on the most recent version, `title_ii_2025.R`. If your office actually intends to follow an older version of the process, every step below could differ. (The unused files are listed, with explanations, at the very end of this document.)
2. **Reporting year.** The most recent script built the report for the cycle covering the academic year Fall 2024 through Summer 2025, with degree completers counted from September 1, 2024 to August 31, 2025. I assumed that "this year's" report follows the same logic with the dates shifted forward to the current reporting year. Every step that names a specific term (Fall, Winter, Spring, Summer) or date range must be updated to the current year's terms and dates. The steps describe terms generically, using the 2025 cycle only as an example.
3. **The code lists are still current.** The script contains fixed lists of teacher education major codes, "initial certification" program codes, and a crosswalk of programs to government codes (reproduced in Appendices A and B). I assumed these lists are still accurate. If any teacher preparation programs have been added, renamed, or discontinued since the last cycle, Steps 6, 12, 15, 16, and 17 would be affected — confirm the lists with the education school or certification officer before starting.
4. **Manual work happened outside the script at several points.** The past analyst stopped several times, fixed things by hand in the saved files, and then continued with the edited file. Where the script's comments explain what was done, it is noted in the step (flagged as "based on a comment in the script"). Where the comments do not explain the work, the step simply flags that something manual happened and names the before/after files so you can compare them. Do not skip these checkpoints.

Title II Report

Date Created: 07-19-2026

Time Created: 11:45:33 PM

R Script Used to Create This Document

- `title_ii_2025.R`

Clarifying Questions

These questions are not crucial to following the steps, but answers would improve this guide:

1. The script's closing comments say an "anticipated completion date" might be needed for enrollees, that a ticket was submitted (on 1/28, "Stuart submitted ticket to Ray") to see if it could be pulled from the student system, and that the plan was to try uploading with it blank. Was that resolved? Should that field be included going forward?
2. Several working files were re-saved with new dates in their names after hand edits (for example, `step_7_title_ii_cohort_filter_20260127.csv` and `title_ii_draft_-`

20260202.csv). Does the office keep notes on what is typically fixed by hand at those points? That would let this guide be more specific at the manual checkpoints.

3. Can the data files be requested with consistent column names from year to year? A large share of the old process was spent renaming columns because each term's file used slightly different names.

A note on using this guide: the steps below are written in plain language, without reference to any particular software, so your new employee can complete them in Excel (or any other tool they prefer).

What This Report Is

Title II refers to a federally required annual report on teacher preparation programs (Title II of the Higher Education Act). The process below produces the institution's **data collection worksheet**: a list of every student who, during the reporting year, was enrolled in or completed one of the institution's teacher preparation programs leading to initial certification, along with the specific data fields the Title II collection requires for each student (identity fields, program codes, years to degree, enrollment status, demographics, and so on).

Data and Inputs Used

All files below are pulled for the **reporting year**, which runs Fall through the following Summer (for the 2025 cycle this was Fall 2024, Winter 2025, Spring 2025, and Summer 2025). The matching is always done using the campus student ID.

1. **End-of-term enrollment files ("EOT"), one per term, for all four terms of the reporting year** (Fall, Winter, Spring, Summer). Fields needed: student ID, SSN, first name, middle initial, last name, date of birth, gender, race/ethnicity (IPEDS — the standardized federal race/ethnicity categories), course level (undergraduate vs. graduate), total credits, higher education history code, and primary award level. (2025-cycle examples: `eot_202490_20250305_final_w_race.csv`, `eot_202510_20250507_final_w_race.csv`, `eot_202520_20260126_final_w_race.csv`, `eot_202560_20260126_final.csv`.)
2. **Student program files ("CSF"), one per term, for the same four terms.** Fields needed: student ID, admit term, catalog term, school/college, department, degree, major, program, and the second school/department/major for double majors. (2025-cycle examples: `student_202490_20250306.csv`, `student_202510_20250306.csv`, `student_eot_20250606.csv`, and `student_list_202560_20250730.csv`.)
3. **Degree completer listing** covering degrees granted from **September 1 through August 31 of the reporting year**. Fields needed: student ID, graduation date, program, degree code, college code, first major code, and degree status code. (2025-cycle example: `Degree Completer Listing_20260127_082131.csv`.)
4. **Official end-of-semester enrollment snapshot files ("ESS") for every past fall and spring term going back at least six years.** These are used to count how many semesters each student has been enrolled, and to fill in identity/demographic details for completers who were not enrolled during the reporting year. Fields needed: student ID, higher education history code, name, date of birth, gender, and race/ethnicity. (The 2025 cycle used twelve files: every fall and spring from Fall 2018 through Spring 2024.)
5. **Reference lists kept inside the old script** (reproduced here in Appendix A and Appendix B): the teacher education major codes, and the initial-certification program list with its government code crosswalk.

Terms and Codes Used Throughout

- **Term codes** are six digits: a four-digit calendar year plus a two-digit term — `10` = Winter, `20` = Spring, `60` = Summer, `90` = Fall. Example: 202490 = Fall 2024; 202520 = Spring 2025.
- **TED** = teacher education. A student is "TED" in a term if their major code (first or second major) that term is on the teacher education list in Appendix A.
- **Initial certification programs** are the subset of teacher education programs that lead to a first (initial) teaching certificate. These are identified by the *program* codes in Appendix B. Only these students belong in the Title II worksheet.
- **Higher education history (HEH) code**: a code on each term's enrollment record describing the student's enrollment-history status. The process treats **1 or 2** as a student's *first semester as an undergraduate*, **4** as a *continuing undergraduate* semester, **6** as a *first semester as a graduate student*, and **7** as a *continuing graduate* semester.
- **CIP code** = Classification of Instructional Programs code, the federal standard code identifying a program's field of study.

A habit worth copying: after nearly every step, the past analyst saved a spreadsheet copy of the work so far (files named `step_1_...`, `step_2_...`, etc.) and eyeballed it before moving on. Save a checkpoint copy after each numbered step below; several later steps reload and build on those saved copies.

Step-by-Step Process

Part One — Build the Year-Long Student List

1. Gather the term files for the reporting year

Collect the four end-of-term enrollment (EOT) files and the four student program (CSF) files — one of each per term (Fall, Winter, Spring, Summer).

2. Add a full-time/part-time status to each term's enrollment file

Create a column in each EOT file marking each student FT or PT that term, based on course level and total credits:

- Undergraduate with 12 or more credits = FT; fewer than 12 = PT.
- Graduate with 9 or more credits = FT; fewer than 9 = PT.
- If course level or credits are missing, leave it blank.

3. Standardize the columns and keep only what is needed

The raw files name the same fields differently from term to term (for example, `LOCAL_ID`, `Campus_ID`, and `id` are all the student ID; the race/ethnicity field appears under many spellings). In every file, rename the fields to one consistent set of names and delete all other columns. Keep:

- *From each EOT file:* student ID, SSN, first name, middle initial, last name, date of birth, gender, IPEDS race/ethnicity, course level, FT/PT, higher education history code, primary award level.
- *From each CSF file:* student ID, admit term, catalog term, school, department, degree, major, program, and second school/department/major.

4. Attach program information to each term's enrollment list

For each term, match the CSF program file to the EOT enrollment file by student ID and bring the program columns over. Keep every student on the enrollment file, whether or not they matched.

5. Clean the major codes

Remove a trailing "W" from the end of any major code (for example, a code ending in `...W` becomes the same code without the W) so the codes will match the master teacher education list.

6. Flag teacher education (TED) majors in each term

In each term's file, add a column marking the student "TED" if their **first major** code is on the Appendix A list, otherwise "NOT." Add a second column doing the same for the **second major**. A blank or missing major counts as "NOT."

7. Combine the four terms into one year-long list

First, label each term's columns with the term they came from (e.g., course level–Fall, course level–Winter, and so on), leaving the identity fields (ID, SSN, name, date of birth, gender, race/ethnicity) unlabeled. Then combine the four term files into a single list with **one row per student, keeping every student who appears in any of the four terms**. After combining:

- For each identity field, keep one final value per student by taking the first non-blank value across the terms.
- Format date of birth as mm/dd/yy. (Watch for dates stored as numbers that have lost a leading zero — an 8-digit date like 09151998 can come in as 9151998 and needs the zero restored before formatting.)
- Treat SSN as text, not a number, so leading zeros are not lost.
- Remove any fully duplicated rows.

Checkpoint saved here in the old process: `step_1_eot_csf_all_terms_all_students_labeled_ted.xlsx`.

8. Keep only students who were TED at some point in the year

Keep each row where at least one of the TED flags (first or second major, any term) says "TED." Also delete any completely blank rows. *(Checkpoint: `step_2_ted_only.xlsx`.)*

9. Flag course-level changes

Add a column showing whether the student's course level (undergraduate/graduate) changed during the year. If the level is the same in every enrolled term, mark "no course lvl change"; otherwise record the term when it first changed. This is used later when reviewing odd cases. *(Checkpoint: `step_3_flag_for_course_level_changes_added.xlsx`.)*

10. Flag whether each student ended the year in a TED major

Add a column marking "yes" if the student's **most recent enrolled term** (Summer if present; otherwise Spring, then Winter, then Fall) shows a TED major. If the student was TED earlier in the year but not in their last term, mark "check" for manual review. Everyone else is "NOT." *(Checkpoint: `step_4_flag_for_last_semester_TED.xlsx`.)*

Part Two — Add Completers and Narrow to the Title II Cohort

11. Add the degree completers

Using the degree completer listing for September 1–August 31 of the reporting year:

- Tidy the column headers (the raw export prefixes each header with database table names — strip those off).

- Keep only: student ID, graduation date, program, degree code, college code, first major code, degree status code.
- Keep only records where the degree status shows the degree was actually **awarded** (status code "UA" or "GA"); drop everything else.
- Remove trailing "W" from the major codes (as in Step 5), then keep only completers whose major is on the teacher education list (Appendix A).
- Merge this completer list into the year-long student list by ID, **keeping everyone from both lists** — this adds completers who were not enrolled during the year.
- Add a graduation status column: "GRADUATED" if the student has a graduation date, otherwise "NOT."

(Checkpoint: `step_5_add_completer_info.xlsx`*.)*

12. Flag students in initial certification programs

Add a column marking "yes" if the student's program is on the initial certification list (Appendix B). Use the program from the student's most recent enrolled term (Summer, else Spring, else Winter, else Fall), and also mark "yes" if the program they *graduated* with is on the list. Everyone else is "NOT." *(Checkpoint:* `step_6_add_flag_for_initial_certs.xlsx`*.)*

13. Narrow to the Title II cohort

- Keep only rows where the initial certification flag is "yes."
- Then identify anyone who was enrolled **only in the fall** (enrolled fall, not spring or summer) **and did not graduate**, and remove them — they left the program without completing and are not reported.

(Checkpoint: `step_7_title_ii_cohort_filter.xlsx`*.)*

14. Manual review checkpoint — check the cohort by hand

Based on comments in the script (not its code), the analyst at this point manually: (a) confirmed no one remained who neither finished the year in a TED major nor graduated with a TED degree; (b) double-checked for duplicate students; and (c) reviewed double majors — that review added two students to the cohort that year. The exact hand edits are not recorded in the script. The file saved in Step 13 was edited and re-saved under a dated name (`step_7_title_ii_cohort_filter_20260127.csv`), and the process continues from the **edited** file. Compare those two files if you want to see exactly what changed last cycle.

Part Three — Build the Title II Data Elements

The final worksheet needs, for each student: last 5 digits of SSN, name, date of birth, program type, reporting group, up to four program codes with CIP codes and degree levels, years to degree, enrollment status, program level, gender, ethnicity, race, program start date, completion date, and institutional program code. The next steps build each of these.

15. Create the identity and grouping fields

- **ssn5:** the last five digits of the SSN. The old process put a letter "X" in front of these digits so spreadsheet software would treat them as text and not chop off leading zeros — the X is removed at the very end (Step 22).
- **program_type:** the letter "T" (for traditional program) for every student. (Caution from the script: some software converts a lone "T" to the logical value TRUE — check for this and change it back if it happens.)
- **reporting_group:** 2 for enrolled students who did not complete; 3 for completers.

- **program_new** (the program to report): for graduates, the program on their degree record; for everyone else, the Summer term program if there is one, otherwise the Spring program. Mark anything that falls through as "CHECK" and resolve by hand.
- **course_level_new:** "UG" if the reported program code starts with B (bachelor's-level), "GR" if it starts with M (master's-level); otherwise "CHECK."

(Checkpoint: `step_8_check_program_course_level.xlsx`.)

16. Assign government program codes, CIP codes, and degree levels

Using the crosswalk in Appendix B, fill in for each student: code 1, CIP code 1, degree 1, and (for programs that report a second code) code 2, CIP code 2, degree 2. Look up the student's **graduation program** first; if the student did not graduate, use the reported enrollment program from Step 15. Degree levels are "B" (bachelor's) or "M" (master's). Enter all codes as **text** so leading zeros and exact decimal forms are kept (e.g., 0126, 13.1). *(Checkpoint: `step_9_check_cip_section.xlsx`.)*

17. Code enrollment status, program level, gender, ethnicity, and race

- **Enrollment status** (1 = full-time, 2 = part-time, 3 = mixed), based on the Fall and Spring FT/PT flags only:
  - FT both Fall and Spring = 1; PT both = 2; one FT and one PT (either order) = 3.
  - Enrolled only one of the two terms: use that term's status (FT = 1, PT = 2).
  - Enrolled neither Fall nor Spring but enrolled in Summer = 2.
  - A comment in the script raises an open question for completers who were not enrolled at all: leave the field blank if the government upload accepts a blank; if not, code them as full-time.
- **Program level:** 1 if degree 1 is "B"; 2 if it is "M."
- **Gender:** M = 1; F = 2.
- **Ethnicity:** Hispanic = 1; any known non-Hispanic race category = 2; nonresident ("NR"), unknown, or anything else = 3.
- **Race:** Black or African American = 1; Asian = 2; American Indian or Alaska Native = 3; Native Hawaiian or Other Pacific Islander = 4; White = 5; Two or more races = 6. Leave race **blank** for Hispanic, nonresident, and unknown.
- **Institutional program code:** look up the program code (last column of Appendix B), again using the graduation program first, otherwise the reported enrollment program.

(Checkpoint: `step_10_crosswalking_title_ii_data_elements.xlsx`.)

Part Four — Semesters Enrolled and Years to Degree

18. Count each student's semesters enrolled

This uses the higher education history (HEH) codes, looking only at **fall and spring** terms, going back six years plus the reporting year itself.

- Gather the twelve past fall/spring end-of-semester snapshot (ESS) files (for the 2025 cycle: Fall 2018 through Spring 2024). Standardize their column names as in Step 3 and keep: ID, HEH code, name, date of birth, gender, and race/ethnicity (the identity fields are kept so they can later fill gaps for completers who were not enrolled during the year).
- Match each snapshot to the cohort by student ID (keeping all cohort rows), labeling each HEH column with its term. Line the HEH columns up in chronological order, ending with the reporting year's own Fall and Spring HEH columns. Consolidate the identity fields as in Step 7 (first non-blank value wins).
- For each student, count semesters according to their course level from Step 15:

– **Undergraduates:** find the term where HEH first equals 1 or 2 (their first semester as an undergraduate), then count every later term where HEH equals 4 (continuing undergraduate), and add 1 for the starting semester.
– **Graduate students:** find the term where HEH first equals 6 (first semester as a graduate student), then count every later term where HEH equals 7 (continuing graduate), and add 1.
– If no starting semester is found in the window, enter "check admit term" — the student likely started more than six years ago or in a summer/winter term, and must be counted by hand.
– If more than one starting semester is found, enter "check" and resolve by hand.

(Checkpoint: `step_11_check_semesters_enrolled_count.xlsx`.)

19. Manual review checkpoint — verify the semester counts

Based on comments in the script (not its code), the analyst at this point manually: verified the counts calculated correctly; filled in or corrected the odd cases flagged "check"/"check admit term" and added notes; added an "x" in front of any codes at risk of losing a leading zero; and back-filled identity/demographic details for completers who were not enrolled during the year. Two additional rules are recorded in the comments: a student admitted in the reporting year's **summer** term counts as **1 semester** enrolled, and such students were counted as 1 year to degree in the prior cycle. The edited file was re-saved under a dated name (`step_11_check_semesters_enrolled_count_20260127.csv`), and the process continues from the **edited** file.

20. Convert semesters enrolled to years to degree

Divide the semester count by two and round up: 1–2 semesters = 1 year; 3–4 = 2; 5–6 = 3; 7–8 = 4; 9–10 = 5; 11–12 = 6. Anything outside 1–12 (including the "check" flags) is marked "enter manually" and must be filled in by hand. *(Checkpoint: `step_12_check_years_to_degree_conversion.xlsx`.)*

21. Work done outside the script — assembling the final workfile

The next stage of the old process starts from a file named `title_ii_workfile_w_notes.csv`, which contains both the coded data elements (Steps 15–17) and the reviewed semesters/years results (Steps 18–20) plus notes. **The script does not show how that combined workfile was assembled** — it appears the analyst brought the pieces together and annotated them by hand. If you need to reconstruct what was done, compare the files:

- *Before:* `step_10_crosswalking_title_ii_data_elements.xlsx` and `step_12_check_years_to_degree_conversion.xlsx`
- *After:* `title_ii_workfile_w_notes.csv`

In practice, the goal of this step is simply: **one table, one row per cohort student, containing every data element built so far.**

Part Five — Final Worksheet Layout and Submission Cleanup

22. Arrange the data into the Title II data collection worksheet layout

- Add empty placeholder columns for a third and fourth program code, CIP code, and degree (the collection template has slots for up to four; this institution only ever fills two).
- Add a **program start date** column (from the admit term — converted to an actual date in Step 23) and a **date of program completion** column (the graduation date; blank for non-completers).
- Keep only the needed columns, in this exact order: student ID; ssn5; last name; first name; middle initial; date of birth; program type; reporting group; code 1; CIP code 1; degree 1; code 2; CIP code

2; degree 2; code 3; CIP code 3; degree 3; code 4; CIP code 4; degree 4; years to degree; enrollment status; program level; gender; ethnicity; race; program start date; date of program completion; program code. (The student ID is kept for internal reference while finalizing.)

(Checkpoint: a dated draft file, e.g. `title_ii_draft_20260128.xlsx`*. Note: in the old process, additional hand edits were made between this draft and a later one — the script next reloads a file dated five days later,* `title_ii_draft_20260202.csv`*, without describing what changed. Compare those files if needed.)*

23. Convert the admit term to a program start date

Turn each student's admit term code into a real date, using the first four digits of the term code as the year and these month/day rules (recorded in the script's comments as the rule provided by a colleague): Fall = September 1; Winter = January 1; Spring = February 1; Summer = June 1. Format as mm/dd/yyyy. Example: admit term 202190 becomes 09/01/2021.

24. Final cleanup just before submission

Per the script's closing comments (not code), do these last:

- Make sure the program type column still says "T" everywhere — fix it if software converted it to TRUE.
- Remove the leading "X"/"x" characters from ssn5 and from the one program code where an x was added to protect a leading zero — but confirm the leading zeros survived.
- Anticipated completion date for enrollees: possibly required; a ticket was pending to see if it could be pulled from the student system, and the plan was to attempt the upload with it blank for non-completers. Check the current collection instructions.

Final Output

One student-level spreadsheet ("the Title II data collection worksheet") with one row per cohort member — every student enrolled in or completing an initial-certification teacher preparation program during the reporting year — containing the 29 columns listed in Step 22, ready to upload to the Title II collection. The process also leaves behind a dated checkpoint file for each step, which is the office's audit trail.

Appendix A — Teacher Education (TED) Major Codes

A student counts as a teacher education major if their first or second major code (after removing any trailing "W") is one of:

ABI, ACH, ACM, AEM, APH, APM, AES, AEN, AFR, AFS, ASP, SST, CHD, ECD, ECI, IEC, TDA, TSD, CSD, SHS, ESL, SLED, HEA, HEC, LTE, PEL, PEM, PHE, SBL, SDBL, SDL

Appendix B — Initial Certification Programs and Code Crosswalk

Only students in these programs belong in the Title II cohort. Enter all codes as **text** (leading zeros and exact decimals matter). "Degree" B = bachelor's, M = master's. A dash means that field is left blank for that program.

Program code(s)	Code 1	CIP 1	Deg. 1	Code 2	CIP 2	Deg. 2	Prog. code
BA_SST (incl. _ANT, _ECO, _GRY, _HIS, _POL, _SOC)	5110	13.1318	B	—	—	—*	22930
BS_ACM	5030	13.1323	B	—	—	—	22922
MAT_ACH	5030	13.1323	M	—	—	—	25108
BS_ABI	5010	13.1322	B	—	—	—	22920
MAT_ABI	5010	13.1322	M	—	—	—	25114
BA_AEN	5013	13.1305	B	—	—	—	22929
MAT_AEN	5013	13.1305	M	—	—	—	25106
BS_AES	5070	13.1337	B	—	—	—	22924
BA_AFS	5140	13.1306	B	5150	13.1306	B	30649
BA_AFR	5140	13.1306	B	—	—	—	22932
BA_ASP	5150	13.1306	B	—	—	—	22931
BA_AEM	5130	13.1311	B	—	—	—	22918
BS_AEM	5130	13.1311	B	—	—	—	22919
BS_APH	5050	13.1329	B	—	—	—	22926
MAT_APH	5050	13.1329	M	—	—	—	25122
BS_APM	5050	13.1329	B	5130	13.1311	B	23377
BS_ECD (incl. _ELA, _EST, _HUM, _MAT, _SOS, _UST)	3013	13.121	B	3014	13.1202	B	23373
BS_ECI (incl. _ELEI, _ESEI, _HMEI, _MTEI, _SSCH, _SSEI)	3013	13.121	B	0126	13.1	B	23374
BS_IEC (incl. _ELA, _EST, _HUM, _MAT, _SOS)	9014	13.1	B	3014	13.1202	B	36629
MST_CHD	3014	13.1202	M	—	—	—	22928
MS_CSD, MS_CSD_CERT	9021	13.1331	M	—	—	—	33410
MST_HEA_NCRT	6121	13.1307	M	—	—	—	25112
BSED_HEC (incl. _CHP, _WEL)	6121	13.1307	B	—	—	—	23375
BSED_PEM (incl. _ADP, _OAE)	6160	13.1314	B	—	—	—	23376
MST_PHE, MST_PHE_PENC	6160	13.1314	M	—	—	—	39290
BA_ESL_CERT	7080	13.14	B	—	—	—	29720

* One inconsistency found in the old script, worth verifying with the education school: for **enrolled (non-completer)** BA_SST students only, the script filled in a second degree level of "B" even though no second code or CIP code is assigned to that program.

Files Provided but Not Used in This Documentation

The following files were provided as input but were **not** used to build this guide:

- **`title_ii_2024.R`** — the previous cycle's (2024) version of this same end-to-end process. It was superseded by `title_ii_2025.R`, which follows the same overall logic with refinements (e.g., computing full-time/part-time status from credits, handling second majors, cleaner name standardization, and added start/completion date fields). Per documentation practice, the more recent version takes priority.

- **Title II Script.R**, **Title II Adding Completers.R**, **Title_II_Script_-Years_Enrolled.R**, and **Years_Enrolled_Add_ins.R** — four fragments of an even earlier (2023) cycle of this report. They respectively: combined the term files into a cohort; matched completers into a state worksheet; looked up past-term enrollment-history codes to determine years enrolled; and repeated that lookup for a handful of late-added students. Everything these scripts did is handled, in updated form, within `title_ii_2025.R`, so they were excluded as outdated versions of the same tasks.